



J. Dairy Sci. TBC

<https://doi.org/10.3168/jds.2025-26949>

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Associations of inflammation measured at 2 DIM with metrics of milk production, health, and herd removal in multiparous Holstein cows

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ABSTRACT

Most dairy cows experience some degree of inflammatory activation during the early postpartum period as a normal response to parturition and the initiation of lactation. Excessive inflammation during this time has been associated with various disease states, impaired reproductive success, and decreased productivity, but time points appropriate for measurement of inflammatory status are not well established. Our objective in this exploratory observational cohort study was to evaluate the relationships between inflammatory markers, specifically haptoglobin and serum amyloid A, measured at 2 DIM and important metrics of lactational success including milk yield through 10 wk in lactation, rumination and activity time during the periparturient period, risk of dyscalcemia, and time to herd removal. Blood was collected at 2 and 4 DIM from 266 multiparous Holstein cows on 4 commercial farms in New York. Serum samples collected at 2 DIM were analyzed for both haptoglobin and serum amyloid A concentrations, and samples from 4 DIM were analyzed for total serum calcium concentrations (tCa). Cows were identified as dyscalcemic if their tCa at 4 DIM was ≤ 2.2 mmol/L. Daily milk weights averaged by week through 10 wk in lactation and herd removal records were collected for each cow. A subset of cows ($n = 101$) had loggers that collected daily rumination times and activities from -14 d relative to parturition to 14 DIM. Linear mixed effects models were used to evaluate the relationships between inflammatory analytes and milk production, rumination time, and activity, and generalized linear models with a quasi-Poisson distribution were used to estimate risk for dyscalcemia. Time to event data for herd removal were modeled using Cox proportional hazards regressions. Confounders including parity, farm, disease status, and calving difficulty, as well as their interactions with analytes and time, were also included in our models where

appropriate. Marginal means, risk ratios, and hazard ratios were calculated from their respective models at 2 concentrations of haptoglobin and serum amyloid A, independently, corresponding to low (haptoglobin: 0.4 g/L; serum amyloid A: 14 μ g/mL), and high (haptoglobin: 1.1 g/L; serum amyloid A: 33 μ g/mL) inflammatory status. Cows with higher inflammatory status produced less milk over the first 9 wk of lactation, ruminated less in the early postpartum period, had decreased activity in the early days following parturition, and had increased risk of dyscalcemia compared with less inflamed cows. Cows with high inflammation were also at a greater hazard of herd removal within 305 DIM. Our data support the hypothesis that more severe postpartum inflammation measured at 2 DIM is associated with reduced lactational success. This evidence suggests that these inflammatory markers might be useful in future diagnostic applications for identifying cows that are at risk of impaired performance in the early postpartum period.

Key words: dyscalcemia, milk yield, rumination, serum amyloid A, haptoglobin

INTRODUCTION

For dairy cows, the early postpartum period is fraught with challenges including energy, protein, and macro-mineral deficits, immunosuppression, inflammatory activation, and microbial invasion of the reproductive tract, among others (Drackley et al., 2005; LeBlanc, 2010; Mezzetti et al., 2021). Though the majority of postparturient cows weather this storm of challenging stimuli and continue through healthy and productive lactations, for some cows, early lactation is marked by the onset of subclinical and clinical disease states (Daros et al., 2022; Gonçalves et al., 2022). The pathophysiological conditions underlying disease development and progression are not fully understood, but once disease processes are initiated, dairy cows that experience them are at increased risk of comorbidities, perform poorly throughout lactation with respect to milk yield and reproductive efficiency, and are typically more likely to be removed

Received May 16, 2025.

Accepted September 26, 2025.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

from the herd (Dubuc et al., 2011; Raboisson et al., 2014; Macmillan et al., 2021). In some cases, these suboptimal lactational outcomes can be predicted through the measurement of certain serum biomarkers within the first 7 DIM (Chapinal et al., 2012; McArt et al., 2012; Caixeta et al., 2017). Therefore, monitoring serum biomarkers in the early postpartum period can be a useful strategy for informing health and management decisions on commercial dairies (Overton et al., 2017).

All cows regardless of health status experience some degree of systemic inflammation during the early postpartum period (Humblet et al., 2006; Trevisi et al., 2012; Fouz et al., 2024). To an extent, inflammation during this timeframe should be expected as a normal response to tissue damage in the reproductive tract and for promoting both expulsion of the placenta and uterine involution (Medzhitov, 2008). However, more severe or long-lasting inflammatory responses are thought to be an indicator that maladaptation is taking place (Bradford et al., 2015; Martins et al., 2021). Low total serum calcium concentrations (**tCa**) during the early postpartum period can also be a useful indicator of maladaptive phenotypes depending on the timing and severity of the drop in calcium (Couto Serrenho et al., 2021; McArt and Oetzel, 2023). Dyscalcemia, a condition characterized by $\text{tCa} \leq 2.2$ mmol/L at 4 DIM, has displayed especially strong potential for use as an indicator of lactational maladaptation due to its associations with increased risk of subsequent disease, decreased milk production, impaired reproductive success, and other physiologically relevant conditions such as depressed rumination time (Neves et al., 2018; Seely and McArt, 2023, 2024). Cows that develop dyscalcemia exhibited elevated concentrations of the positive acute-phase proteins serum amyloid A and haptoglobin 2 d preceding dyscalcemia diagnosis, but it remains unclear how concentrations of inflammatory markers measured at 2 DIM may be associated with negative outcomes (Seminara et al., 2025). Early identification of maladaptive phenotypes at 2 DIM, through associations with inflammatory markers, could be useful in the development of treatment or management strategies.

Our prospective cohort study aimed to evaluate the associations between inflammatory analytes (haptoglobin and serum amyloid A) measured at 2 DIM and important metrics of successful postpartum adaptation. These metrics included weekly milk yields through 10 wk in lactation, daily periparturient rumination and activity times from 14 d prepartum to 14 DIM, risk of dyscalcemia at 4 DIM, and hazard of herd removal through 305 DIM. We hypothesized that increased concentrations of both analytes would be associated with suboptimal metric outcomes.

MATERIALS AND METHODS

The STROBE-Vet reporting guidelines were implemented in both the design and analysis of our exploratory prospective cohort study (Sargeant et al., 2016).

Study Population, Sample Collection and Sample Analysis

Our study was conducted using serum samples banked from a previous field trial (Seely et al., 2023). Due to the exploratory nature of this study, we did not conduct an independent sample size calculation, opting instead to use all available samples. The original randomized control trial enrolled all eligible cows entering their second or greater lactation at 2 farms in northeastern New York from June to July 2021 (farms A and B) and at 2 farms in central New York from July to September 2021 (farms C and D). Farms were eligible for inclusion in the study if they milked at least 1,000 Holstein cows 3 times daily, had headlocks in fresh cow pens, fed a negative dietary cation anion difference diet prepartum and a TMR postpartum (diet compositions reported in Seely et al., 2023), and if the farms recorded daily milk weights using DairyComp 305 (Valley Agricultural Software, Tulare, CA). Blood samples were collected around the time of morning feed delivery at 2 and 4 DIM from coccygeal vessels into evacuated 10-mL anticoagulant-free tubes using 20-gauge vacutainer needles (Becton Dickinson). Following clotting and centrifugation, serum was separated, aliquoted, and stored at -20°C until analysis. All available cows from the control group, who received no postpartum calcium supplementation, were retrospectively enrolled into our analysis ($n = 267$).

Concentration of serum amyloid A at 2 DIM was assessed using commercially available kits on an automated analyzer (Hitachi Modular P800, Roche Diagnostics) at the New York State Animal Health Diagnostic Center (Ithaca, NY), with a range of quantification from 5 $\mu\text{g/mL}$ to 200 $\mu\text{g/mL}$. The inter- and intra-assay coefficients of variation for serum amyloid A analysis were 3.2% and 2.3%, respectively. Haptoglobin concentrations at 2 DIM were assessed on an automated analyzer (Cobas c501, Roche Diagnostics) at the University of Guelph Animal Health Laboratory (Guelph, ON, Canada), using a method based on Makimura and Suzuki (1982). Inter- and intra-assay coefficients of variation for haptoglobin analysis were 14.1% and 1.4%, respectively. Total serum calcium analysis was conducted on 4 DIM serum samples using commercially available kits (Calcium Gen.2, Roche Diagnostics, Indianapolis, IN) on an automated analyzer (Hitachi Modular P800, Roche Diagnostics) at the New York State Animal Health and Diagnostic Center (Ithaca,

NY), with an interassay CV of 1.1% and an intra-assay CV of 0.9%. Cows with $tCa \leq 2.2$ mmol/L were classified as dyscalcemic.

Herd removal events (sold or died), milk production, calving difficulty (defined as having twins or stillbirth), and early postpartum disease occurrence (diagnosed with milk fever, displaced abomasum, metritis, or removed from the herd at <5 DIM) determined by farm personnel were recorded using DairyComp 305 (Valley Agricultural Software, Tulare, CA). Daily milk yields were averaged on a weekly basis over the first 10 wk of lactation.

A subset of cows on farms C and D were equipped with validated rumination and activity loggers (SCR Dairy, Netanya, Israel; Schirmann et al., 2009). On farm C, neck collar loggers were used ($n = 44$), and farm D used ear tag loggers ($n = 57$). Rumination and activity times were recorded every 2 h and automatically transferred to system software (Dataflow; SCR Dairy) using antennae located within barns, every 20 min. All 2-h records were summed for each 24-h period between 14 d prepartum and 14 DIM, yielding daily rumination times (min/d) and daily activity (arbitrary units/d).

Statistical Analysis

Calculations of descriptive statistics and all statistical analyses were performed using R version 4.4.0 in RStudio (Posit Software, PBC, Boston, MA). General functions contained in the base and stats packages of R were used to generate descriptive tables and plots and to determine correlations. Continuous variables were summarized as means and standard deviations, medians and interquartile range, minimum, and maximum, and the distributions of haptoglobin and serum amyloid A were described using histograms. Categorical variables were summarized as counts and percentages.

Linear mixed effects models, provided in the lme4 package (Bates et al., 2015), were used to evaluate the average daily milk yield over 10 wk in lactation, as well as rumination and activity times from 14 d prepartum to 14 DIM. For both rumination and activity time data pre- and postpartum responses were modeled separately. Milk production and postpartum rumination and activity times were modeled using the same strategy: The response was modeled as a function of the concentration of the selected analyte (haptoglobin or serum amyloid A) as a continuous variable, time (week of lactation in milk analyses and DIM in rumination and activity analyses) as a categorical variable, and the interaction between the analyte and time. Four potential categorical confounders at the population level, parity group (2, 3, and ≥ 4), farm, early postpartum disease status, calving difficulty (twins or stillbirth), were also included in every model, as fixed effects. In addition to the main predictors of interest (an-

alyte, time, and their interaction) and the potential confounding variables listed above included in every model regardless of significance, the following terms were all tested and removed if $P > 0.1$, using a backward stepwise process to improve model parsimony: the 2-way interactions of the analyte term with parity group and farm; the quadratic term for each analyte and its 2-way interactions with time, parity group and farm; the 2-way interaction between farm and time; the 2-way interaction between parity group and time; the 3-way interactions between the analyte, time and parity group or farm; the 3-way interactions between the quadratic term for each analyte, time and parity group or farm. In milk yield models, the 2-way interactions of the analyte and its quadratic term with calving difficulties, and the 3-way interactions of these terms with time, were also tested. The 2- and 3-way interactions involving early postpartum disease status, the analyte terms, and time were not evaluated due to limited early disease incidence.

A similar model fitting approach was applied to prepartum rumination and activity time data. In prepartum models, responses were modeled as a function of the selected analyte as a continuous variable, time (d relative to calving) as a categorical variable, the interaction between the analyte and time, and the categorical confounders parity group and farm. Additionally, the following terms were all tested and removed if $P > 0.1$, using a backward stepwise process: the 2-way interactions of the analyte term with parity group and farm; the quadratic term for each analyte and its 2-way interactions with time, parity group and farm; the 2-way interaction between farm and time; the 2-way interaction between parity group and time; the 3-way interactions between the analyte, time, and parity group or farm; the 3-way interactions between the quadratic terms for each analyte, time, and parity group or farm. Because concentrations of inflammatory analytes were determined postpartum, retrospective relationships with prepartum rumination and activity responses should be interpreted descriptively; no causality can be inferred from prepartum models.

For all linear mixed effects models, the random effect of cow was included to account for correlations between repeated measures taken on the same individual over time. By default, the lme4 package assumes a compound symmetric structure within the random effect covariance matrix (Gurka et al., 2011; Bates et al., 2015). Model assumptions were assessed through visual inspection of diagnostic plots, residuals plotted over the range of predicted values to evaluate the homoscedasticity assumption, and histograms of residuals to evaluate their normality. All model assumptions were satisfactorily met. Proportionally weighted marginal means and 95% CI were calculated at each time point using the emmeans

package (Lenth, 2024). Pairwise comparisons were conducted between predicted means at each time point.

To evaluate the relationship between our analytes and risk of dyscalcemia, we fit generalized linear models using a quasi-Poisson distribution. Each model was fit using the same strategy, including the continuous concentration of a single analyte and the categorical variables parity group, farm, calving difficulty, and disease status. The quadratic term for each analyte and the interactions of the analyte and its quadratic term with parity group, farm, and calving difficulty were also included if $P \leq 0.1$. A 95% CI of the risk of dyscalcemia was calculated over the range of relevant inflammatory marker concentrations, and risk ratios (RR) with 95% CI for comparisons between inflammatory thresholds, described below, were calculated for dyscalcemia using the emmeans package (Lenth, 2024).

Cox proportional hazards models, from the survival package of R (Therneau, 2024), were constructed to evaluate the association of analyte concentration with hazard of herd removal within 305 d of lactation. The outcome of interest in this analysis was days to removal from the herd predicted as a function of analyte concentration, parity group, farm, calving difficulty, and early postpartum disease status. The quadratic term for analyte concentration and the interactions between the analyte terms and parity group, farm, and calving difficulty were included in each respective model if $P \leq 0.1$. Cows were right censored at 305 DIM. Models were fit for each analyte individually by the same backward stepwise model fitting protocol, and the fit was assessed by checking the proportional hazards assumption, which was met in both models. Hazard ratios (HR) were calculated using the emmeans package (Lenth, 2024).

After a parsimonious model was fit for each response, with inflammatory analytes included as continuous predictors, each model was used to calculate proportionally weighted marginal means, RR, and HR at 2 reference concentrations of each analyte, independently. The reference concentration thresholds selected for model calculations were chosen a priori to represent either low (LO; haptoglobin: 0.4 g/L; serum amyloid A: 14 µg/mL) or high (HI; haptoglobin: 1.1 g/L; serum amyloid A: 33 µg/mL) postpartum inflammatory activation. Retrospective classification of sampled cows based on these thresholds was not performed, rather the responses reported are the model-based predictions at specified analyte values with all other covariates set to proportionally weighted average values. Threshold concentrations of serum amyloid A and haptoglobin corresponding to different degrees of postpartum inflammation are not established at 2 DIM or any day in lactation, therefore the thresholds selected in this study were based on relevant adjacent research. For haptoglobin, the findings of Huzzey et al. (2009) showed

that cows without metritis at 3 DIM had serum concentrations of haptoglobin around 0.4 g/L. Seminara et al. (2025) found that healthy cows with normal calcium status also had serum concentrations of haptoglobin at 0.4 g/L at 2 DIM. Those same authors showed independently that cows with metritis or dyscalcemia had haptoglobin concentrations of 1.1 g/L at 2 and 3 DIM. As a result, we chose these threshold concentrations for comparison. For serum amyloid A, previously published data were sparse. Though there are a few studies that have investigated serum amyloid A at early postpartum time points, under varying inflammatory contexts, these studies have found that cows experienced concentrations of serum amyloid A well out of the range of the concentrations of serum amyloid A detected in this population at 2 DIM (Chan et al., 2010; Chandler et al., 2022; Couto Serrenho et al., 2023). Based on Seminara et al. (2025) the LO and HI inflammation concentrations thresholds were selected to be 14 µg/mL and 33 µg/mL, matching the concentrations of serum amyloid A in the blood of eucalcemic cows and dyscalcemic cows, respectively at 2 DIM.

RESULTS

Descriptive Characteristics

The original cohort study from which we sampled had 267 cows with banked serum at 2 DIM and calcium data at 4 DIM. Of these, one cow was excluded due to excessive hemolysis in its 2 DIM sample, resulting in a sample size of 266 multiparous Holsteins for the final analysis. Means and distributional characteristics of continuous variables are in Table 1. The breakdown of cows by parity group and farm; the incidences of calving difficulties, early postpartum disease, dyscalcemia, and herd removal within 305 DIM; and the number of cows included in the rumination and activity data analyses are in Table 2. The right-skewed distributions of haptoglobin and serum amyloid A for the study population are in Figure 1; the Pearson correlation between these 2 variables at 2 DIM was 0.77.

Milk Yield

Summaries of the final models used to calculate average daily milk yield over the first 10 wk of lactation from haptoglobin and serum amyloid A are in Table 3. Average daily milk yield over the first 10 wk of lactation, calculated from haptoglobin at LO and HI inflammatory statuses at 2 DIM for each parity group are in Figure 2. Figure 3 represents the calculations from a model incorporating serum amyloid A as the predictor.

Haptoglobin. The interactions between both the haptoglobin term, and its quadratic term, with parity group and

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Table 1. Descriptive statistics of continuous variables measured in a cohort of 266 multiparous Holsteins on 4 farms in northern and central New York, including haptoglobin and serum amyloid A measured at 2 DIM, average daily milk yield over the first 10 wk of lactation, and rumination and activity times¹

Variable	Mean (±SD)	Median (interquartile range)	Minimum	Maximum
Haptoglobin, g/L	0.7 (±0.6)	0.5 (0.3–0.8)	0.1	3.6
Serum amyloid A, µg/mL	20.1 (±16.9)	14.8 (8.6–25.7)	<0.5	105
Average milk yield, kg/d	49.8 (±9.3)	50.9 (44.5–55.9)	8.2	73.6
Rumination time, min/d				
Prepartum	516 (±87)	518 (462–575)	39	779
Postpartum	494 (±125)	524 (430–584)	24	713
Activity, AU ² /d				
Prepartum	393 (±67)	390 (348–431)	201	644
Postpartum	411 (±75)	407 (363–451)	161	794

¹Measured from –14 d relative to parturition, to parturition, and from the day of parturition through 14 DIM.²AU = arbitrary units.

time, revealed that the association of 2 DIM haptoglobin with milk yield was dependent on the week of lactation and the parity of the cow. We therefore stratified model calculations for each haptoglobin threshold over time by parity group (Figure 2). Calculations from the linear models revealed that parity 2 cows at LO inflammatory status produced more milk over the first 3 wk of lactation than cows with HI inflammatory status (Figure 2A). Differences between the yields of cows with HI and LO inflammatory statuses in parity 2 were largest during the second week of lactation, in which LO cows produced 42.4 kg/d (40.8 to 44.0 kg/d), and HI cows produced 38.2 kg/d (35.8 to 40.7 kg/d; $P = 0.002$). In parity 3, the LO cows produced more milk than HI cows through 9 wk of lactation (all $P \leq 0.1$; Figure 2B). During wk 2 the difference between LO and HI cows was the largest, with LO cows producing 44.7 kg/d (42.8 to 46.6 kg/d), whereas HI cows produced 37.5 kg/d (34.2 to 40.8 kg/d; $P < 0.001$). Cows with parity ≥ 4 showed a similar pattern whereby yield was greater for cows with LO inflammatory status at 1 through 6 wk of lactation and then again at 8 and 9 wk of lactation (all $P \leq 0.1$; Figure 2C). The difference between yields for parity ≥ 4 cows with LO and HI inflammation was the greatest during the third week of

lactation, when LO cows produced 9.3 kg of milk more on average each day (mean [95% CI]; LO: 45.5 [43.5 to 47.4]; HI: 36.2 [32.3 to 40.0]; $P < 0.001$).

Serum Amyloid A. As there were no statistical interactions between serum amyloid A concentrations and parity group in the final model, results for milk yield with serum amyloid A as the primary predictor were presented as single set of marginal means proportionally weighted across farm and parity group. Over the first 10 wk of lactation, calculations from the linear models revealed that cows with LO inflammatory status, relative to serum amyloid A, produced more milk than cows with HI inflammatory statuses for the first 9 wk in lactation (all $P \leq 0.1$; Figure 3). Despite the significance of this trend, the numerical differences in milk yield after the third week of lactation were not large. The largest difference between the milk yields of LO and HI threshold cows was detected at 2 wk of lactation when LO cows produced 41.9 kg/d (40.9 to 43.0 kg/d) and HI cows produced 37.6 kg/d (36.3 to 38.9 kg/d; $P < 0.001$).

Table 2. Descriptive information for categorical variables in a cohort of 266 multiparous Holsteins sampled on 4 farms in northern and central New York, including cows by farm and parity group, as well as numbers of cows with dyscalcemia (tCa ≤ 2.2 mmol/L at 4 DIM), cows removed from the herd within 305 DIM, and cows included in rumination and activity time analysis

Farm	Parity group				Health confounder		Health outcome		Rumination and activity, n (%)
	Total	2, n (%)	3, n (%)	≥ 4 , n (%)	Calving difficulty, ¹ n (%)	Disease status, ² n (%)	Dyscalcemia, n (%)	Removal, ³ n (%)	
A	69	23 (33)	22 (32)	24 (35)	5 (7)	0 (0)	16 (23)	19 (28)	—
B	33	20 (61)	6 (18)	7 (21)	3 (9)	0 (0)	6 (18)	8 (24)	—
C	44	17 (39)	13 (30)	14 (31)	3 (7)	0 (0)	15 (34)	4 (9)	44 (37)
D	120	60 (50)	33 (29)	27 (21)	8 (7)	7 (6)	29 (24)	20 (17)	57 (48)
Total	266	120 (45)	74 (28)	72 (27)	19 (7)	7 (3)	66 (25)	51 (19)	101 (38)

¹Defined as having twins or stillbirth.²Defined as being diagnosed with displaced abomasum, metritis, or being removed from the herd before 5 DIM.³Herd removal within 305 DIM.

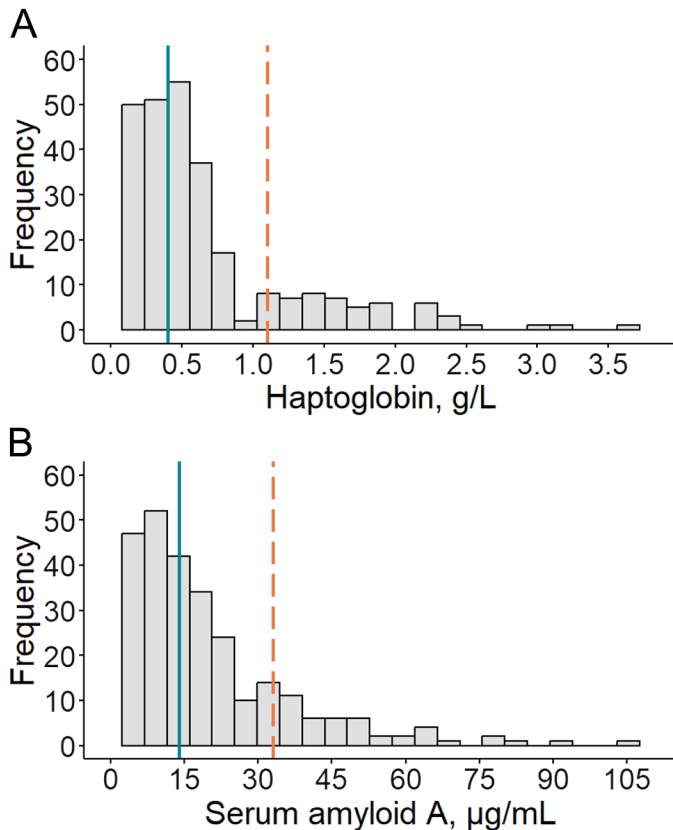


Figure 1. Histograms displaying the distributions of haptoglobin (A) and serum amyloid A (B) concentrations in the blood of 266 multiparous Holsteins on 4 farms in northern and central New York, measured at 2 DIM. Lines indicate low and high inflammatory threshold concentrations used for comparisons in all analyses. Solid turquoise corresponds to the low inflammatory threshold (LO: haptoglobin = 0.4 g/L, serum amyloid A = 14 µg/mL), whereas dashed orange corresponds to the high inflammatory threshold (HI: haptoglobin = 1.1 g/L, serum amyloid A = 33 µg/mL).

Rumination and Activity Time

Summaries of the final models used to predict pre- and postpartum rumination and activity times are included in Table 3. Figures 4 and 5 show the patterns of periparturient rumination times and activity times by day, respectively, from 14 d prepartum through 14 DIM calculated at LO and HI inflammatory statuses, by parity group. Panels A, B, and C represent models incorporating haptoglobin as the main predictor, while panels D, E, and F represent models incorporating serum amyloid A as the main predictor.

Haptoglobin and Rumination Time. Overall, as seen in Figure 4, we observed only mild differences between inflammatory statuses during the prepartum period, most notably in cows with parity ≥ 4 . Postpartum, cows with LO inflammatory status in parity 2 had increased daily rumination times compared with HI cows, at 2, 3, 4, and 5 DIM (all $P < 0.05$; Figure 4A). Cows with LO inflam-

matory status in parity 3 had increased daily rumination times compared with HI cows from the day of calving through 10 DIM (all $P < 0.05$; Figure 4B). Cows with LO inflammatory status in parities ≥ 4 did not have notable differences in postpartum rumination times, only showing elevated rumination time at 8 DIM.

Serum Amyloid A and Rumination Time. Differences in rumination times prepartum were unremarkable when calculated from the model at LO and HI inflammatory statuses (Figure 4). Postpartum, parity 2 cows at the LO inflammatory threshold had higher daily rumination times than cows at the HI inflammatory threshold at 3, 4, and 5 DIM (all $P < 0.05$; Figure 4D), whereas parity 3 cows at the LO inflammatory threshold had elevated rumination times from 2 through 12 DIM compared with cows at the HI inflammatory threshold (all $P < 0.05$; Figure 4E). Parity ≥ 4 cows at the LO inflammatory threshold had higher daily rumination times from 2 through 5 DIM (all $P < 0.05$; Figure 4F).

Haptoglobin and Activity. Activity times calculated from haptoglobin-based models showed minimal prepartum differences. Parity 2 cows at the LO inflammatory threshold had slightly elevated activity compared with HI cows in the 6 d preceding parturition (all $P \leq 0.1$; Figure 5A) but these differences were not apparent for parity 3 and ≥ 4 cows. Postpartum there were continuous differences between activity times for parity 2 and ≥ 4 cows persisting through 11 DIM, whereby cows at the LO inflammatory threshold were more active than cows at the HI inflammatory threshold (all $P \leq 0.1$; Figure 5A and 5C). Parity 3 cows had more minor differences, whereby cows at the LO inflammatory threshold were more active than HI cows only on 2 and 3 DIM (all $P < 0.05$; Figure 5B).

Serum Amyloid A and Activity. The calculations from the prepartum model revealed that cows with LO inflammatory status at 2 DIM relative to serum amyloid A were somewhat more active for 6 d preceding parturition than cows at the HI inflammatory threshold (all $P \leq 0.1$; Figure 5D–F). The postpartum model showed that parity 2 cows at the LO inflammatory threshold were more active than parity 2 cows at the HI threshold from 2 through 11 DIM (all $P \leq 0.1$; Figure 5D). Parity 3 cows at the LO threshold were more active than HI cows from 1 through 4 DIM (all $P \leq 0.05$; Figure 5E), and parity ≥ 4 cows at the LO threshold were only more active than HI cows on the day of calving and 1 DIM (all $P \leq 0.05$; Figure 5F).

Dyscalcemia

Summaries of the final models used to predict risk of dyscalcemia from haptoglobin and serum amyloid A are in Table 3. Though significant interactions were detected between both analytes and calving difficulty, the direc-

Table 3. Summaries of final model fit (*P*-values) for models used to assess the associations between 2 DIM concentrations of haptoglobin (HP) and serum amyloid A (SAA), and various metrics of lactational success in a cohort of 266 multiparous Holsteins on 4 farms in northern and central New York

Item	Milk yield ¹			Rumination time ²			Activity time ³			Dyscemia ⁴		Herd removal within 305 DIM ⁵	
	HP		SAA	Pre	Post	HP	HP	Post	HP	HP	SAA	HP	SAA
	HP	SAA	HP	Pre	Post	HP	HP	Post	HP	SAA	HP	HP	SAA
Analyte	0.006	<0.001	0.6	<0.001	0.01	0.01	0.5	<0.001	<0.001	0.003	<0.001	0.06	0.2
Analyte ²	0.1	—	0.95	0.008	0.002	0.5	—	0.01	<0.001	0.004	<0.001	0.06	0.04
Time	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.1	<0.001	<0.001	<0.001	—	—	—
Analyte ² × time	0.002	<0.001	0.01	0.04	0.3	<0.001	0.002	0.05	—	0.2	—	—	—
Analyte × time	0.4	—	<0.001	0.8	0.2	0.2	—	<0.001	<0.001	0.3	—	—	—
Parity group	0.2	0.1	0.03	0.6	0.02	0.5	0.02	0.004	0.005	<0.001	0.4	0.02	0.02
Analyte × parity	0.2	—	0.006	0.4	0.1	0.09	0.1	0.06	—	0.8	—	—	0.04
Analyte ² × parity	0.1	—	0.001	0.6	0.03	0.09	—	—	—	—	—	—	0.01
Parity × time	<0.001	<0.001	0.07	<0.001	—	<0.001	0.1	0.3	—	0.006	—	—	—
Analyte × parity × time	0.005	—	<0.001	0.1	—	<0.001	0.09	0.006	—	0.05	—	—	—
Analyte ² × parity × time	0.04	—	<0.001	0.08	—	<0.001	—	—	—	—	—	—	—
Farm	<0.001	<0.001	<0.001	0.05	<0.001	0.5	<0.001	<0.001	<0.001	<0.001	0.03	0.2	0.2
Analyte × farm	0.003	0.08	—	0.9	0.01	0.3	—	—	—	0.4	—	—	0.03
Analyte ² × farm	—	—	—	0.9	—	—	—	—	—	0.5	—	—	—
Farm × time	<0.001	<0.001	—	<0.001	—	<0.001	<0.001	0.002	<0.001	<0.001	—	—	—
Analyte × farm × time	—	—	—	0.004	—	<0.001	—	—	—	0.01	—	—	—
Analyte ² × farm × time	—	—	—	0.02	—	—	—	—	—	0.02	—	—	—
Calving difficulty	0.9	0.7	—	0.3	—	0.2	—	0.9	—	0.9	0.5	0.3	0.1
Analyte × CD ⁶	0.005	—	—	0.2	—	0.9	—	0.3	—	0.6	0.04	0.03	0.02
Analyte ² × CD	0.01	—	—	0.06	—	—	—	0.07	—	0.2	—	0.03	0.07
CD × time	—	—	—	0.6	—	0.5	—	<0.001	<0.001	<0.001	—	—	—
Analyte × CD × time	—	—	—	0.004	—	0.02	—	0.05	—	<0.001	—	—	—
Analyte ² × CD × time	—	—	—	<0.001	—	—	—	0.04	—	<0.001	—	—	—
Disease status ⁷	0.6	0.7	—	0.4	—	0.5	—	0.05	—	0.4	0.3	0.7	0.1

¹Average daily milk yield over the first 10 wk of lactation, using a linear mixed effects model.²Prepartum (Pre) and postpartum (Post) rumination times, predicted using a linear mixed effects model.³Prepartum and postpartum activity times, predicted using a linear mixed effects model.⁴Risk of dyscemia from HP and SAA, predicted using a quasi-Poisson model.⁵Herd removal within 305 DIM, predicted using Cox proportional hazards.⁶CD = calving difficulty, defined as having twins or stillbirth.⁷Defined as being diagnosed with displaced abomasum, or metritis, or being removed from the herd before 5 DIM.

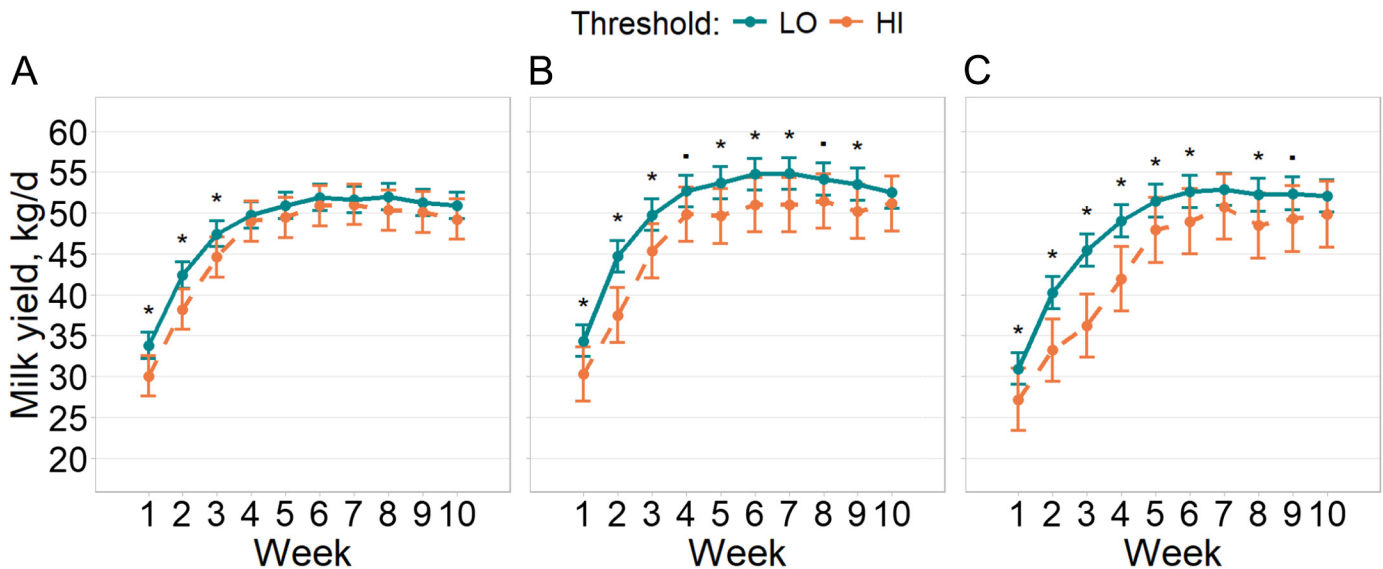


Figure 2. Average daily milk yield across the first 10 wk of lactation stratified by parity (A: parity 2; B: parity 3; C: parity ≥ 4) and calculated at 2 concentrations of haptoglobin in the blood at 2 DIM, corresponding to LO (0.4 g/L), and HI (1.1 g/L) inflammatory status, for a cohort of 266 multiparous Holsteins on 4 farms in northern and central New York. The dots (•) indicate $0.05 < P \leq 0.1$ for the pairwise comparison; asterisks (*) indicate differences ($P \leq 0.05$) between calculated means. Error bars represent 95% CI.

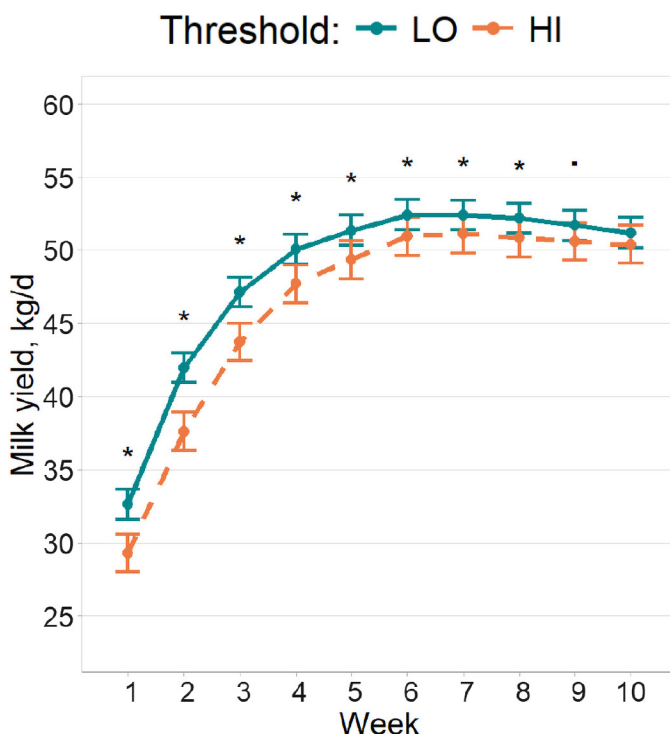


Figure 3. Average daily milk yield across the first 10 wk of lactation calculated at 2 concentrations of serum amyloid A in the blood at 2 DIM, corresponding to LO (14 $\mu\text{g/mL}$), or HI (33 $\mu\text{g/mL}$) inflammatory status, from a cohort of 266 multiparous Holsteins on 4 farms in northern and central New York. The dots (•) indicate $0.05 < P \leq 0.1$ for the pairwise comparison; asterisks (*) indicate differences ($P \leq 0.05$) between calculated means. Error bars represent 95% CI.

tion of the association was not different between cows with and without calving difficulty for either predictor.

Haptoglobin. The curve describing risk of dyscalcemia over a relevant range of haptoglobin concentrations is in Figure 6A, showing that as haptoglobin concentrations increase, risk of dyscalcemia becomes greater, but to a diminished extent at higher concentrations of haptoglobin. When this same model was used to predict RR between inflammatory status groups, it was found that HI cows were at an elevated risk of dyscalcemia compared with LO cows (RR = 4.0, 95% CI = 2.5 to 6.1; $P < 0.001$).

Serum Amyloid A. The curve describing risk of dyscalcemia over a relevant range of serum amyloid A concentrations is in Figure 6B. Similar to haptoglobin, the risk of dyscalcemia increased with increasing serum amyloid A concentrations. The RR comparing the risk of dyscalcemia between LO and HI inflammatory status groups, based on serum amyloid A revealed that HI cows had elevated risk of dyscalcemia compared with LO cows (RR = 2.5, 95% CI = 1.8–3.5; $P < 0.001$).

Time to Herd Removal Within 305 DIM

Summaries of the final models used to calculate hazard of herd removal before 305 DIM are in Table 3. When these models were used to calculate HR comparing LO and HI inflammatory status groups based on haptoglobin concentrations, results suggested that HI cows were at increased hazard of herd removal within 305 DIM compared with LO cows (HR = 1.8, 95% CI = 1.0–3.2;

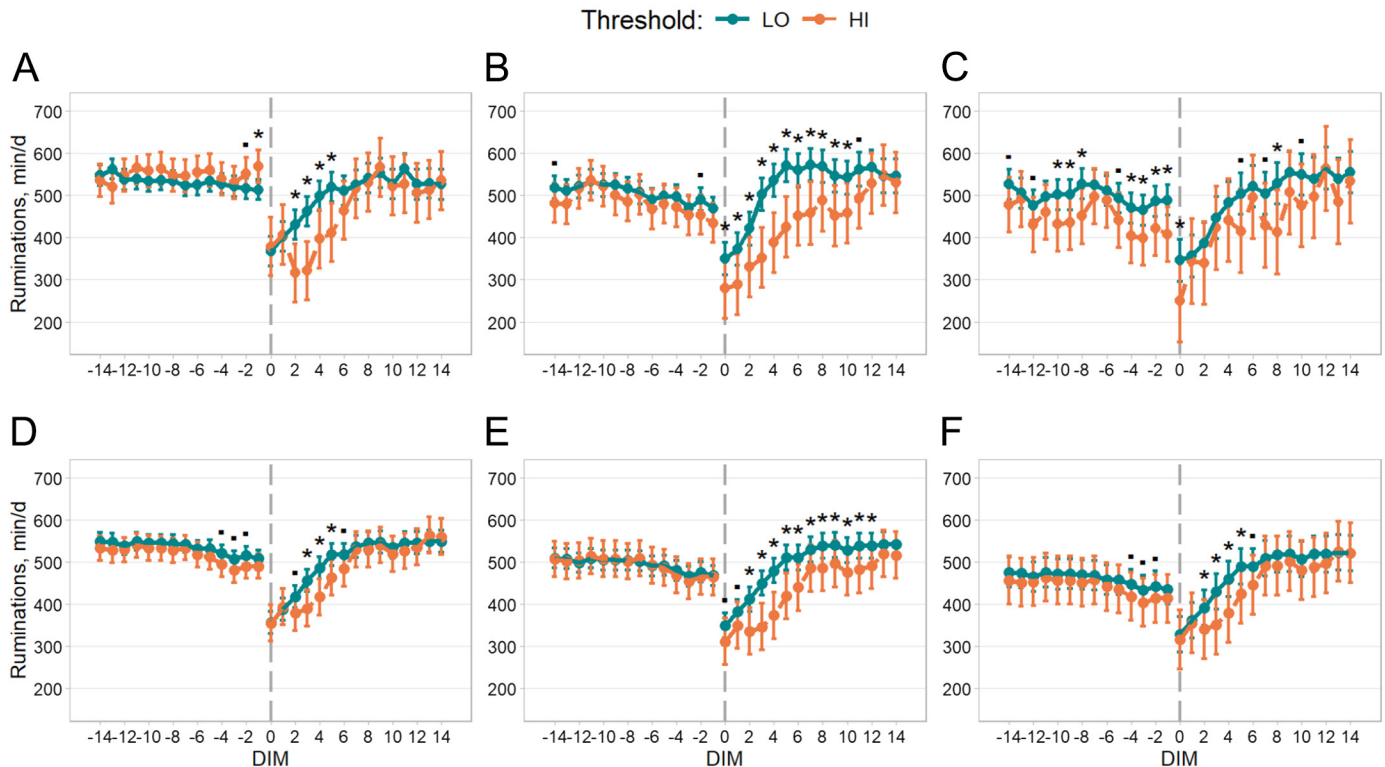


Figure 4. Rumination times (min/d) for a cohort of 101 multiparous Holstein cows sampled on 2 farms in central New York, stratified by parity group, with LO and HI inflammatory status at 2 DIM assessed using haptoglobin (A: parity 2, B: parity 3, and C: parity ≥ 4) and serum amyloid A (D: parity 2, E: parity 3, and F: parity ≥ 4) concentrations in the blood (LO: haptoglobin = 0.4 g/L, serum amyloid A = 14 $\mu\text{g/mL}$; HI: haptoglobin = 1.1 g/L, serum amyloid A = 33 $\mu\text{g/mL}$). The dots (•) indicate $0.05 < P \leq 0.1$ for the pairwise comparison; asterisks (*) indicate differences ($P \leq 0.05$) between calculated means. Error bars represent 95% CI.

$P = 0.07$). When accounting for several significant interactions, hazard ratios comparing inflammatory status thresholds based on serum amyloid A concentrations showed that hazard of herd removal was increased for HI cows compared with LO cows (HR = 2.3, 95% CI = 1.1–4.8; $P = 0.03$). Interactions suggested that parity, farm, and calving difficulties were all important in determining the magnitude of this relationship although the direction of the association was not changed across the strata of these confounders.

DISCUSSION

Our objective in this exploratory study was to evaluate the associations between 2 positive acute-phase inflammatory analytes, haptoglobin and serum amyloid A, measured at 2 DIM and important metrics of early lactation success and postpartum adaptation. We hypothesized that higher concentrations of these analytes would be associated with more negative outcomes. Our findings with respect to milk production, rumination and activity times, and risk of dyscalcemia supported our hypothesis. Findings around herd removal showed some suggestive asso-

ciations that indicate we may have been underpowered to assess those outcomes and further studies are needed to more rigorously evaluate that relationship. Overall, these data suggest that inflammation measured as early as 2 DIM has important negative implications for lactational performance. For the sake of brevity, this discussion will focus mainly on haptoglobin, as there is greater body of literature concerning bovine haptoglobin with which to compare and contrast.

Inflammation, identified by a variety of physiologically relevant markers, has been known to be associated with depressed milk production for many years (Bertoni et al., 2008; Trevisi et al., 2015). Induced inflammation models suggest that inflammatory activation may be the causal agent driving these losses, although the mechanisms mediating this relationship remain to be elucidated (Horst et al., 2020; Opgenorth et al., 2024a,c). Haptoglobin, specifically, has received attention in this regard, and it has been consistently shown that greater haptoglobin concentrations, measured at a range of early postpartum time points, are associated with sustained losses in milk production (Huzzey et al., 2015; McCarthy et al., 2016; Kerwin et al., 2022). Our findings support this body of

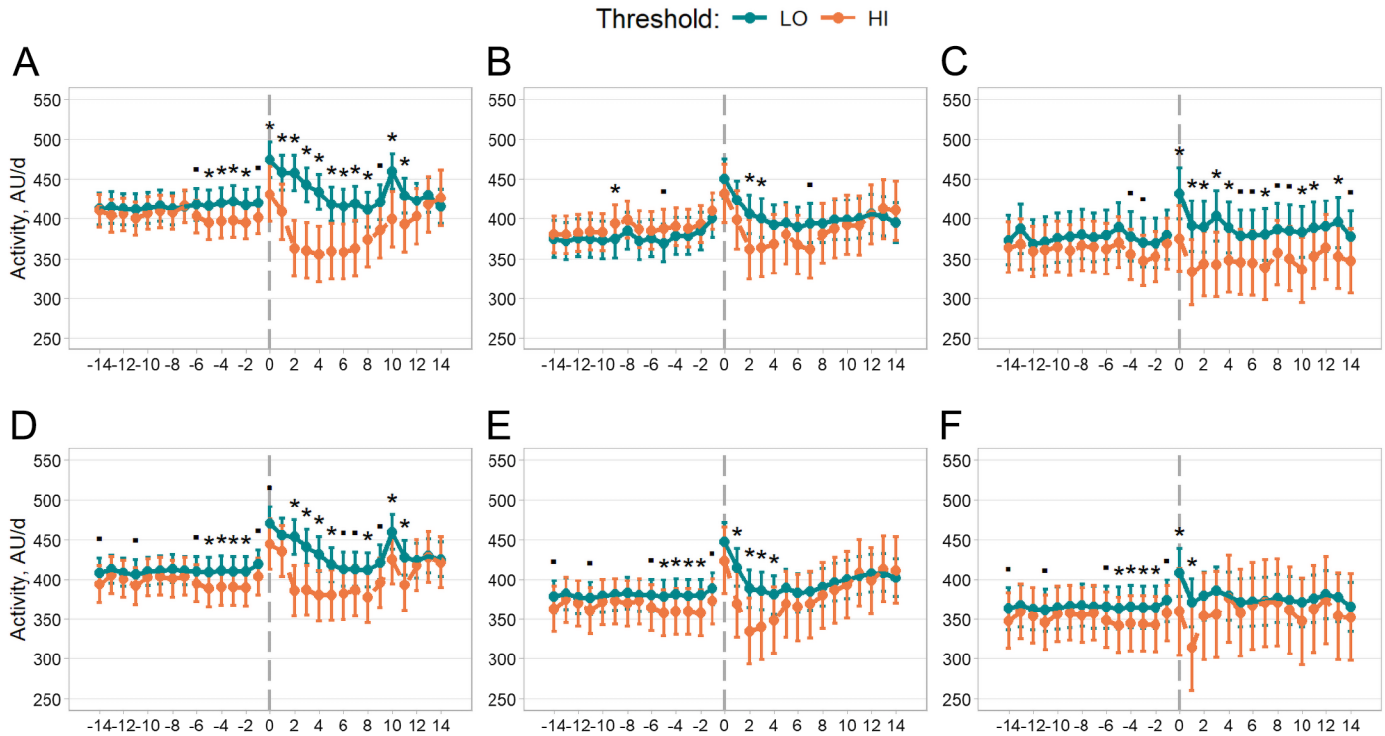


Figure 5. Activity times (arbitrary units [AU]/d) for a cohort of 101 multiparous Holstein cows sampled on 2 farms in central New York, stratified by parity group, with LO and HI inflammatory status at 2 DIM assessed using haptoglobin (A: parity 2, B: parity 3, and C: parity ≥ 4) and serum amyloid A (D: parity 2, E: parity 3, and F: parity ≥ 4) concentrations in the blood (LO: haptoglobin = 0.4 g/L, serum amyloid A = 14 $\mu\text{g/mL}$; HI: haptoglobin = 1.1 g/L, serum amyloid A = 33 $\mu\text{g/mL}$). The dots (•) indicate $0.05 < P \leq 0.1$ for the pairwise comparison; asterisks (*) indicate differences ($P \leq 0.05$) between calculated means. Error bars represent 95% CI.

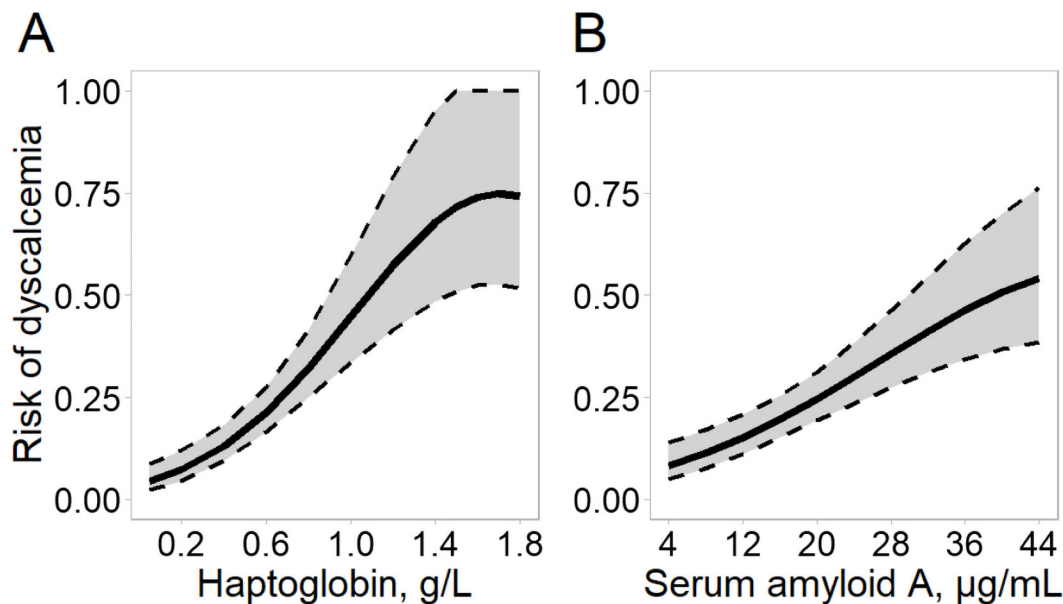


Figure 6. Risk of dyscalcemia (tCa ≤ 2.2 mmol/L at 4 DIM) over a range of haptoglobin (A) and serum amyloid A (B) concentrations in the blood of multiparous Holstein cows at 2 DIM predicted from a cohort of 266 cow sampled on 4 farms in northern and central New York. Values calculated from models including the analytes, their quadratic terms, parity group (2, 3, and ≥ 4), farm, calving difficulty (twins or stillborn), early postpartum disease status, and all relevant 2-way interactions. Gray areas bounded by dashed lines represent a 95% CI.

literature, adding that measurements of haptoglobin as early as 2 DIM can be informative about sustained productivity deficits.

Rumination and activity time have been consistently associated with health status, such that cows with disease have reduced rumination and activity times in the early postpartum period and around the time of disease diagnosis (Schirmann et al., 2016; Barragan et al., 2018; Cocco et al., 2021). Stangaferro et al. (2016a,b) found that cows with various diseases, including hyperketonemia, metritis, and displaced abomasum, had depressed rumination times around their time of diagnosis, and these same cows experienced elevated concentrations of haptoglobin in the blood at 3 DIM. Diagnosed disease occurrence in this cohort was minimal; therefore, it is difficult to determine how much of the inflammation we observed was related to underlying pathology, but based on the above evidence, it seems likely that the early development of undiagnosed disease processes might have been a factor driving the relationships we observed. Based on the statistical relevance of calving difficulty in our models, it also seems likely that issues during parturition contributed to the relationships we observed. Though calving ease scores were not reliably assessed on our farms, it is known that cows with difficult calvings also experience proinflammatory phenotypes in the early postpartum period and have been shown to have a pattern of periparturient rumination times similar to what we observed in this study (Mammi et al., 2021). In light of these preliminary findings, the biological machinery connecting rumination, activity, disease, calving difficulty, and early postpartum inflammation remains fertile grounds for future research.

It is well known that inflammation and calcium status are associated. Models of induced inflammation have demonstrated that acute inflammatory stimuli cause serum tCa to drop (Horst et al., 2020; Chandler et al., 2022; Opgenorth et al., 2024b), but data illustrating the extent of this relationship under naturally occurring postpartum conditions are lacking. Recent exploratory work by Couto Serrenho et al. (2025) has shown that lower tCa at 2 DIM are associated with increased haptoglobin concentrations in clinically healthy cows, but that this relationship is variable at different time points within the first week of lactation. In an exploratory case-control study, Seminara et al. (2025) showed that clinically healthy cows with dyscalcemia had elevated concentrations of haptoglobin at 2 DIM compared with cows with normal calcium status at 4 DIM. Though both recent studies support the hypothesis that calcium status and inflammatory activation are connected in the early postpartum period under naturally occurring conditions, neither study was designed to establish the directionality of this relationship. Our findings that increased haptoglobin concentrations at 2 DIM

were strongly associated with risk of dyscalcemia add to this narrative linking calcium regulation and inflammation in the early postpartum period.

Our findings around herd removal were slightly more ambiguous. Though the time to herd removal analysis did not reach significance with pairwise comparisons for haptoglobin, the effect sizes of the HR between HI cows and LO cows, based on both analytes were highly suggestive that HI cows are removed from the herd at increased rates compared with LO cows. These findings are not surprising given that haptoglobin concentrations have been associated with increased risk of herd removal within 30 DIM previously (Kerwin et al., 2022). Furthermore, Wilson et al. (2022) found that hazard of herd removal over multiple lactations was reduced for cows that received a postpartum anti-inflammatory therapy, a finding that suggests the inflammation itself may be a central driver of this maladaptive phenotype. Research aimed at clarifying this relationship and its implications with respect to welfare and management using a more appropriate sample size is warranted.

In many of our analyses, parity group interactions proved to be important, showing that cows of different parities were impacted by inflammation differentially. It is known that multiparous ruminants have distinct metabolic physiology compared with their primiparous counterparts during the postpartum period and both groups experience different degrees of inflammation at this time (Schneider et al., 2013; Rupprechter et al., 2018; Cabiddu et al., 2020). Furthermore, there is evidence that multiparous cows generally experience lower levels of inflammatory activation compared with primiparous cows during the postpartum period, but that they may be more sensitive to elevated concentrations of inflammatory mediators in their blood (Burfeind et al., 2014; McCarthy et al., 2016). Our findings with respect to milk production push this concept further, suggesting that ascending parities within the multiparous category may become less tolerant of more excessive inflammation as they age.

It is not obvious why haptoglobin and serum amyloid A displayed different relationships with our outcomes of interest despite their strong correlation, although in the postpartum context the distinct biological functions of these molecules might hold a potential explanation. Serum amyloid A has a wide variety of functions, including but not limited to cholesterol scavenging, opsonization of pathogens, and modulation of innate immune reactions (Ceciliani et al., 2012; Sack, 2018). Haptoglobin also has number of functions, but arguably the most important is to bind free hemoglobin and prevent oxidative stress (Ceciliani et al., 2012; Bertaggia et al., 2014). This function might be of particular relevance at a time in a dairy cow's productive cycle when excessive oxidative

stress is a major concern and thought to be a driver of negative outcomes (Sordillo and Aitken, 2009; Abuelo et al., 2014). Based on this evidence, it is possible that oxidative stress is a factor in lactational maladaptation, contributing to the relationships we observed, but future research will have to test this hypothesis with a different study design.

Limitations

There are a few important limitations to recognize in this study. First, this was an exploratory analysis; the parent study was not designed to assess these outcomes. It is, therefore, difficult to determine whether there were not differences in our outcomes between cows at different inflammatory thresholds, or we were merely underpowered to detect them. The design of this study and its associated sample size concerns certainly limited the outcomes we were able to accurately assess. This study took place during one season of one year, therefore we cannot establish whether these associations exist under different seasonal conditions, or if the associations are repeatable from year to year. With our rumination and activity data in particular, different sensor types were used on different farms; therefore, there is the possibility that different performance of these sensors could have influenced our results. That said, our inclusion of farm-level variables in the models may well have accounted for possible differences in sensor performance. Evaluating these relationships only in multiparous cows also limits the applicability of this study's findings to the overall lactating population. The threshold concentrations we used to predict LO and HI inflammatory status are also a key limitation, as previous research has not clearly established these thresholds as having biological importance. Furthermore, disease diagnosis may have been performed more effectively on some farms than others therefore under- or over-reporting is possible, and it is not totally clear to what extent we were able to control the confounding effect of these processes. Another limitation worth recognizing is the fact that this study is purely associative, in that we cannot draw causal conclusions from the relationships that we have observed.

In spite of these limitations, the associations we observed demonstrated important long-lasting differences between cows of varying inflammatory status at 2 DIM, whereby cows with higher inflammation generally had reduced milk production, reduced postpartum rumination and activity times, increased risk of dyscalcemia, and increased hazard of herd removal. It is possible that this knowledge could be leveraged in the development of diagnostic strategies, which could in turn influence intervention decisions aimed at mitigating these negative

associations, although much larger and sufficiently powered studies would be warranted before these outcomes could be achieved. Parsing out the biological underpinnings of these relationships to establish their causal directionality is the critical next step for determining what interventions might be effective.

CONCLUSIONS

Elevated inflammation measured via serum concentrations of haptoglobin and, to a lesser extent, serum amyloid A at 2 DIM are useful indicators that cows are likely to produce less milk across the first 10 wk of lactation, ruminate less during the early postpartum period, be less active during the initial days following calving, and have an increased risk of developing dyscalcemia. Parity appears to be important when considering early postpartum elevations in inflammation. Overall, our findings that long-term metrics of lactational success are negatively associated with inflammatory activation at 2 DIM support the concept that early postpartum inflammation is connected to lactational maladaptation.

NOTES

This work was supported by the American Association of Bovine Practitioners Foundation in a sponsored agreement with Cornell University (OSP#147873), as well as the Animal Health and Production and Animal Products-Improved Nutritional Performance, Growth, and Lactation of Animals grant no. 2019-67015-29446/project accession no. 1018890 from the USDA National Institute of Food and Agriculture (Washington, DC). The authors would like to thank the participating farms, Clair Seely (University of New Hampshire, Durham, NH) for her leadership in the parent study from which our samples were collected, and Stephen Parry (Cornell University Statistical Consulting Unit, Ithaca, NY) for his expert statistical guidance. Data used in our study were from cows sampled via a protocol approved by the Cornell University Institutional Animal Care and Use Committee (protocol number 2020-0102). Jessica McArt serves as the editor in chief of *JDS Communications*. The authors have not stated any other conflicts of interest.

Nonstandard abbreviations used: AU = arbitrary units; HI = inflammatory status threshold corresponding to high inflammation; HP = haptoglobin; HR = hazard ratio; LO = inflammatory status threshold corresponding to low inflammation; Pre = prepartum; Post = postpartum; RR = risk ratio; SAA = serum amyloid A; tCa = total serum calcium concentration.

REFERENCES

- Abuelo, A., J. Hernández, J. L. Benedito, and C. Castillo. 2015. The importance of the oxidative status of dairy cattle in the periparturient period: Revisiting antioxidant supplementation. *J. Anim. Physiol. Anim. Nutr. (Berl.)* 99:1003–1016. <https://doi.org/10.1111/jpn.12273>.
- Barragan, A. A., J. M. Piñeiro, G. M. Schuenemann, P. J. Rajala-Schultz, D. E. Sanders, J. Lakritz, and S. Bas. 2018. Assessment of daily activity patterns and biomarkers of pain, inflammation, and stress in lactating dairy cows diagnosed with clinical metritis. *J. Dairy Sci.* 101:8248–8258. <https://doi.org/10.3168/jds.2018-14510>.
- Bates, D., M. Mächler, B. Bolker, and S. Walker. 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67:1–48. <https://doi.org/10.18637/jss.v067.i01>.
- Bertagaglia, E., G. Scabia, S. Dalise, F. lo Verso, F. Santini, P. Vitti, C. Chisari, M. Sandri, and M. Maffei. 2015. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* 67:1–48. <https://doi.org/10.18637/jss.v067.i01>.
- Bertoni, G., E. Trevisi, X. Han, and M. Bionaz. 2008. Effects of inflammatory conditions on liver activity in puerperium period and consequences for performance in dairy cows. *J. Dairy Sci.* 91:3300–3310. <https://doi.org/10.3168/jds.2008-0995>.
- Bradford, B. J., K. Yuan, J. K. Farney, L. K. Mamedova, and A. J. Carpenter. 2015. Invited review: Inflammation during the transition to lactation: New adventures with an old flame. *J. Dairy Sci.* 98:6631–6650. <https://doi.org/10.3168/jds.2015-9683>.
- Burfeind, O., I. Sannmann, R. Voigtsberger, and W. Heuwieser. 2014. Receiver operating characteristic curve analysis to determine the diagnostic performance of serum haptoglobin concentration for the diagnosis of acute puerperal metritis in dairy cows. *Anim. Reprod. Sci.* 149:145–151. <https://doi.org/10.1016/j.anireprosci.2014.07.020>.
- Cabiddu, A., M. Dattena, M. Decandia, G. Molle, V. Lopreiato, A. Minuti, and E. Trevisi. 2020. The effect of parity number on the metabolism, inflammation, and oxidative status of dairy sheep during the transition period. *J. Dairy Sci.* 103:8564–8575. <https://doi.org/10.3168/jds.2019-18114>.
- Caixeta, L. S., P. A. Ospina, M. B. Capel, and D. Nydam. 2017. Association between subclinical hypocalcemia in the first 3 days of lactation and reproductive performance of dairy cows. *Theriogenology* 94:1–7. <https://doi.org/10.1016/j.theriogenology.2017.01.039>.
- Cecilian, F., J. J. Ceron, P. D. Eckersall, and H. Sauerwein. 2012. Acute phase proteins in ruminants. *J. Proteomics* 75:4207–4231. <https://doi.org/10.1016/j.jprot.2012.04.004>.
- Chan, J. P. W., C. C. Chang, W. L. Hsu, W. B. Liu, and T. H. Chen. 2010. Association of increased serum acute-phase protein concentrations with reproductive performance in dairy cows with postpartum metritis. *Vet. Clin. Pathol.* 39:72–78. <https://doi.org/10.1111/j.1939-165X.2009.00182.x>.
- Chandler, T. L., T. A. Westhoff, A. S. Sipka, T. R. Overton, and S. Mann. 2022. Lipopolysaccharide challenge following intravenous amino acid infusion in postpartum dairy cows: II. Clinical and inflammatory responses. *J. Dairy Sci.* 105:4611–4623. <https://doi.org/10.3168/jds.2021-21227>.
- Chapinal, N., M. E. Carson, S. J. LeBlanc, K. E. Leslie, S. Godden, M. Capel, J. E. P. Santos, M. W. Overton, and T. F. Duffield. 2012. The association of serum metabolites in the transition period with milk production and early-lactation reproductive performance. *J. Dairy Sci.* 95:1301–1309. <https://doi.org/10.3168/jds.2011-4724>.
- Cocco, R., M. E. A. Canozzi, and V. Fischer. 2021. Rumination time as an early predictor of metritis and subclinical ketosis in dairy cows at the beginning of lactation: Systematic review-meta-analysis. *Prev. Vet. Med.* 189:105309. <https://doi.org/10.1016/j.prevetmed.2021.105309>.
- Couto Serrenho, R., T. J. DeVries, T. F. Duffield, and S. J. LeBlanc. 2021. Graduate Student Literature Review: What do we know about the effects of clinical and subclinical hypocalcemia on health and performance of dairy cows? *J. Dairy Sci.* 104:6304–6326. <https://doi.org/10.3168/jds.2020-19371>.
- Couto Serrenho, R., E. Morrison, T. C. Bruinje, and S. J. LeBlanc. 2023. Assessment of systemic inflammation following oral calcium supplementation in healthy postpartum multiparous dairy cows—A randomized controlled trial. *JDS Commun.* 5:134–138. <https://doi.org/10.3168/jdsc.2023-0438>.
- Couto Serrenho, R., R. C. Neves, and S. J. LeBlanc. 2025. Association between postpartum systemic inflammation and serum calcium in healthy multiparous dairy cows—An exploratory observational analysis. *JDS Commun.* <https://doi.org/10.3168/jdsc.2024-0725>. In press.
- Daros, R. R., D. M. Weary, and M. A. G. von Keyserlingk. 2022. Invited review: Risk factors for transition period disease in intensive grazing and housed dairy cattle. *J. Dairy Sci.* 105:4734–4748. <https://doi.org/10.3168/jds.2021-20649>.
- Drackley, J. K., H. M. Dann, G. N. Douglas, N. A. Janovick Guretzky, N. B. Litherland, J. P. Underwood, and J. J. Loo. 2005. Physiological and pathological adaptations in dairy cows that may increase susceptibility to periparturient diseases and disorders. *Ital. J. Anim. Sci.* 4:323–344. <https://doi.org/10.4081/ijas.2005.323>.
- Dubuc, J., T. F. Duffield, K. E. Leslie, J. S. Walton, and S. J. LeBlanc. 2011. Effects of postpartum uterine diseases on milk production and culling in dairy cows. *J. Dairy Sci.* 94:1339–1346. <https://doi.org/10.3168/jds.2010-3758>.
- Fouz, R., R. Rodríguez-Bermúdez, I. J. Rodríguez-Godina, M. Rodríguez-Domínguez, M. Rico, and F. J. Diéguez. 2024. Evaluation of haptoglobin concentration in clinically healthy dairy cows: Correlation between serum and milk levels. *J. Appl. Anim. Res.* 52:2300624. <https://doi.org/10.1080/09712119.2023.2300624>.
- Gonçalves, J. L., J. L. de Campos, A. J. Steinberger, N. Safdar, A. Kates, A. Sethi, J. Shutske, G. Suen, T. Goldberg, R. I. Cue, and P. L. Ruegg. 2022. Incidence and treatments of bovine mastitis and other diseases on 37 dairy farms in Wisconsin. *Pathogens* 11:1282. <https://doi.org/10.3390/pathogens11111282>.
- Horst, E. A., E. J. Mayorga, M. Al-Qaisi, M. A. Abeyta, S. L. Portner, C. S. McCarthy, B. M. Goetz, S. K. Kvidera, and L. H. Baumgard. 2020. Effects of maintaining eucalcemia following immunoactivation in lactating Holstein dairy cows. *J. Dairy Sci.* 103:7472–7486. <https://doi.org/10.3168/jds.2020-18268>.
- Humblet, M. F., H. Guyot, B. Boudry, F. Mbayahi, C. Hanzen, F. Rollin, and J. M. Godeau. 2006. Relationship between haptoglobin, serum amyloid A, and clinical status in a survey of dairy herds during a 6-month period. *Vet. Clin. Pathol.* 35:188–193. <https://doi.org/10.1111/j.1939-165X.2006.tb00112.x>.
- Huzzey, J. M., T. F. Duffield, S. J. LeBlanc, D. M. Veira, D. M. Weary, and M. A. G. von Keyserlingk. 2009. Short communication: Haptoglobin as an early indicator of metritis. *J. Dairy Sci.* 92:621–625. <https://doi.org/10.3168/jds.2008-1526>.
- Huzzey, J. M., S. Mann, D. Nydam, R. J. Grant, and T. R. Overton. 2015. Associations of peripartum markers of stress and inflammation with milk yield and reproductive performance in Holstein dairy cows. *Prev. Vet. Med.* 120:291–297. <https://doi.org/10.1016/j.prevetmed.2015.04.011>.
- Kerwin, A. L., W. S. Burhans, S. Mann, D. Nydam, S. K. Wall, K. M. Schoenberg, K. L. Perfield, and T. R. Overton. 2022. Transition cow nutrition and management strategies of dairy herds in the northeastern United States: Part II—Associations of metabolic- and inflammation-related analytes with health, milk yield, and reproduction. *J. Dairy Sci.* 105:5349–5369. <https://doi.org/10.3168/jds.2021-20863>.
- LeBlanc, S. 2010. Monitoring metabolic health of dairy cattle in the transition period. *J. Reprod. Dev.* 56(Suppl):S29–S35. <https://doi.org/10.1262/jrd.1056S29>.
- Lenth, R. 2024. emmeans: Estimated marginal means, aka least squares means. R package version 1.10.1. Accessed XXX. <https://CRAN.R-project.org/package=emmeans>.
- Macmillan, K., M. Gobikrushanth, A. Behrouzi, B. Hoff, and M. G. Colazo. 2021. Prevalence of early postpartum health disorders in Holstein cows and associations with production, reproduction, and survival outcomes on Alberta dairy farms. *Can. Vet. J.* 62:273–280.
- Mammi, L. M. E., D. Cavallini, M. Fustini, I. Fusaro, M. Giammarco, A. Formigoni, and A. Palmonari. 2021. Calving difficulty influences

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- rumination time and inflammatory profile in Holstein dairy cows. *J. Dairy Sci.* 104:750–761. <https://doi.org/10.3168/jds.2020-18867>.
- Martins, L. F., P. R. Menta, L. Fernandes, V. S. Machado, and R. C. Neves. 2021. Prolonged, low-grade inflammation in the first week of lactation: Associations with mineral, protein, and energy balance markers, and milk yield, in a clinically healthy Jersey cow cohort. *J. Dairy Sci.* 104:6113–6123. <https://doi.org/10.3168/jds.2020-19538>.
- McArt, J. A. A., D. Nydam, and G. R. Oetzel. 2012. Epidemiology of subclinical ketosis in early lactation dairy cattle. *J. Dairy Sci.* 95:5056–5066. <https://doi.org/10.3168/jds.2012-5443>.
- McArt, J. A. A., and G. R. Oetzel. 2023. Considerations in the diagnosis and treatment of early lactation calcium disturbances. *Vet. Clin. North Am. Food Anim. Pract.* 39:241–259. <https://doi.org/10.1016/j.cvfa.2023.02.009>.
- McCarthy, M. M., T. Yasui, M. J. B. Felipe, and T. R. Overton. 2016. Associations between the degree of early lactation inflammation and performance, metabolism, and immune function in dairy cows. *J. Dairy Sci.* 99:680–700. <https://doi.org/10.3168/jds.2015-9694>.
- Medzhitov, R. 2008. Origin and physiological roles of inflammation. *Nature* 454:428–435. <https://doi.org/10.1038/nature07201>.
- Mezzetti, M., L. Cattaneo, M. M. Passamonti, V. Lopreato, A. Minuti, and E. Trevisi. 2021. The transition period updated: A review of the new insights into the adaptation of dairy cows to the new lactation. *Dairy*. 2:617–636. <https://doi.org/10.3390/dairy2040048>.
- Neves, R. C., B. M. Leno, K. D. Bach, and J. A. A. McArt. 2018. Epidemiology of subclinical hypocalcemia in early lactation Holstein dairy cows: The temporal associations of plasma calcium concentration in the first 4 days in milk with disease and milk production. *J. Dairy Sci.* 101:9321–9331. <https://doi.org/10.3168/jds.2018-14587>.
- Opgenorth, J., M. A. Abeyta, B. M. Goetz, S. Rodriguez-Jimenez, A. D. Freestone, R. P. Rhoads, R. P. McMillan, J. L. McGill, and L. H. Baumgard. 2024a. Intramammary lipopolysaccharide challenge in early versus mid-lactation dairy cattle: Immune, production, and metabolic responses. *J. Dairy Sci.* 107:6252–6267. <https://doi.org/10.3168/jds.2023-24488>.
- Opgenorth, J., E. J. Mayorga, M. A. Abeyta, B. M. Goetz, S. Rodriguez-Jimenez, A. D. Freestone, J. L. McGill, and L. H. Baumgard. 2024b. Intravenous lipopolysaccharide challenge in early versus mid-lactation dairy cattle. I: The immune and inflammatory responses. *J. Dairy Sci.* 107:6225–6239. <https://doi.org/10.3168/jds.2023-24350>.
- Opgenorth, J., E. J. Mayorga, M. A. Abeyta, S. Rodriguez-Jimenez, B. M. Goetz, A. D. Freestone, and L. H. Baumgard. 2024c. Intravenous lipopolysaccharide challenge in early- versus mid-lactation dairy cattle. II: The production and metabolic responses. *J. Dairy Sci.* 107:6240–6251. <https://doi.org/10.3168/jds.2023-24351>.
- Overton, T. R., J. A. A. McArt, and D. Nydam. 2017. A 100-Year Review: Metabolic health indicators and management of dairy cattle. *J. Dairy Sci.* 100:10398–10417. <https://doi.org/10.3168/jds.2017-13054>.
- Raboisson, D., M. Mounié, and E. Maigné. 2014. Diseases, reproductive performance, and changes in milk production associated with subclinical ketosis in dairy cows: A meta-analysis and review. *J. Dairy Sci.* 97:7547–7563. <https://doi.org/10.3168/jds.2014-8237>.
- Rupprechter, G., M. de L. Adrien, A. Larriestra, O. Meotti, C. Batista, A. Meikle, and M. Noro. 2018. Metabolic predictors of peri-partum diseases and their association with parity in dairy cows. *Res. Vet. Sci.* 118:191–198. <https://doi.org/10.1016/j.rvsc.2018.02.005>.
- Sack, G. H. Jr. 2018. Serum amyloid A—A review. *Mol. Med.* 24:46. <https://doi.org/10.1186/s10020-018-0047-0>.
- Sargeant, J. M., A. M. O'Connor, I. R. Dohoo, H. N. Erb, M. Cevallos, M. Egger, A. K. Ersbøll, S. W. Martin, L. R. Nielsen, D. L. Pearl, D. U. Pfeiffer, J. Sanchez, M. E. Torrence, H. Vigre, C. Waldner, and M. P. Ward. 2016. Methods and processes of developing the strengthening the reporting of observational studies in epidemiology—Veterinary (STROBE-Vet) statement. *Prev. Vet. Med.* 134:188–196. <https://doi.org/10.1016/j.prevetmed.2016.09.005>.
- Schirmann, K., M. A. G. von Keyserlingk, D. M. Weary, D. M. Veira, and W. Heuwieser. 2009. Validation of a system for monitoring rumination in dairy cows. *J. Dairy Sci.* 92:6052–6055. <https://doi.org/10.3168/jds.2009-2361>.
- Schirmann, K., D. M. Weary, W. Heuwieser, N. Chapinal, R. L. A. Cerri, and M. A. G. von Keyserlingk. 2016. Short communication: Rumination and feeding behaviors differ between healthy and sick dairy cows during the transition period. *J. Dairy Sci.* 99:9917–9924. <https://doi.org/10.3168/jds.2015-10548>.
- Schneider, A., M. N. Corrêa, and W. R. Butler. 2013. Short communication: Acute phase proteins in Holstein cows diagnosed with uterine infection. *Res. Vet. Sci.* 95:269–271. <https://doi.org/10.1016/j.rvsc.2013.02.010>.
- Seely, C. R., and J. A. A. McArt. 2023. The association of subclinical hypocalcemia at 4 days in milk with reproductive outcomes in multiparous Holstein cows. *JDS Commun.* 4:111–115. <https://doi.org/10.3168/jdsc.2022-0279>.
- Seely, C. R., and J. A. A. McArt. 2024. Patterns of periparturient rumination and activity time in multiparous Holstein cows with and without dyscalcemia in early lactation. *J. Dairy Sci.* 107:4871–4880. <https://doi.org/10.3168/jds.2023-24139>.
- Seely, C. R., C. N. Wilbur, K. M. Fang, and J. A. A. McArt. 2024. Effects of timing of oral calcium administration on milk production in high producing early lactation Holstein cows. *J. Dairy Sci.* 107:1620–1629. <https://doi.org/10.3168/jds.2023-23541>.
- Seminara, J. A., C. R. Seely, and J. A. A. McArt. 2025. Acute phase responses in clinically healthy multiparous Holsteins with and without calcium dysregulation during the early postpartum period. *J. Dairy Sci.* 108:1930–1939. <https://doi.org/10.3168/jds.2024-25300>.
- Sordillo, L. M., and S. L. Aitken. 2009. Impact of oxidative stress on the health and immune function of dairy cattle. *Vet. Immunol. Immunopathol.* 128:104–109. <https://doi.org/10.1016/j.vetimm.2008.10.305>.
- Stangaferro, M. L., R. Wijma, L. S. Caixeta, M. A. Al-Abri, and J. O. Giordano. 2016a. Use of rumination and activity monitoring for the identification of dairy cows with health disorders: Part I. Metabolic and digestive disorders. *J. Dairy Sci.* 99:7395–7410. <https://doi.org/10.3168/jds.2016-10907>.
- Stangaferro, M. L., R. Wijma, L. S. Caixeta, M. A. Al-Abri, and J. O. Giordano. 2016b. Use of rumination and activity monitoring for the identification of dairy cows with health disorders: Part III. Metritis. *J. Dairy Sci.* 99:7422–7433. <https://doi.org/10.3168/jds.2016-11352>.
- Therneau, T. 2024. A package for survival analysis in R. R package version 3.8–3. Accessed XXX. <https://CRAN.R-project.org/package=survival>.
- Trevisi, E., M. Amadori, S. Cogrossi, E. Razzuoli, and G. Bertoni. 2012. Metabolic stress and inflammatory response in high-yielding, periparturient dairy cows. *Res. Vet. Sci.* 93:695–704. <https://doi.org/10.1016/j.rvsc.2011.11.008>.
- Trevisi, E., N. Jahan, G. Bertoni, A. Ferrari, and A. Minuti. 2015. Pro-inflammatory cytokine profile in dairy cows: Consequences for new lactation. *Ital. J. Anim. Sci.* 14:3862. <https://doi.org/10.4081/ijas.2015.3862>.
- Wilson, J. P., M. J. Green, L. Randall, C. S. Rutland, N. J. Bell, H. Hemingway-Arnold, J. S. Thompson, N. J. Bollard, and J. N. Huxley. 2022. Effects of routine treatment with nonsteroidal anti-inflammatory drugs at calving and when lame on the future probability of lameness and culling in dairy cows: A randomized controlled trial. *J. Dairy Sci.* 105:6041–6054. <https://doi.org/10.3168/jds.2021-21329>.

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